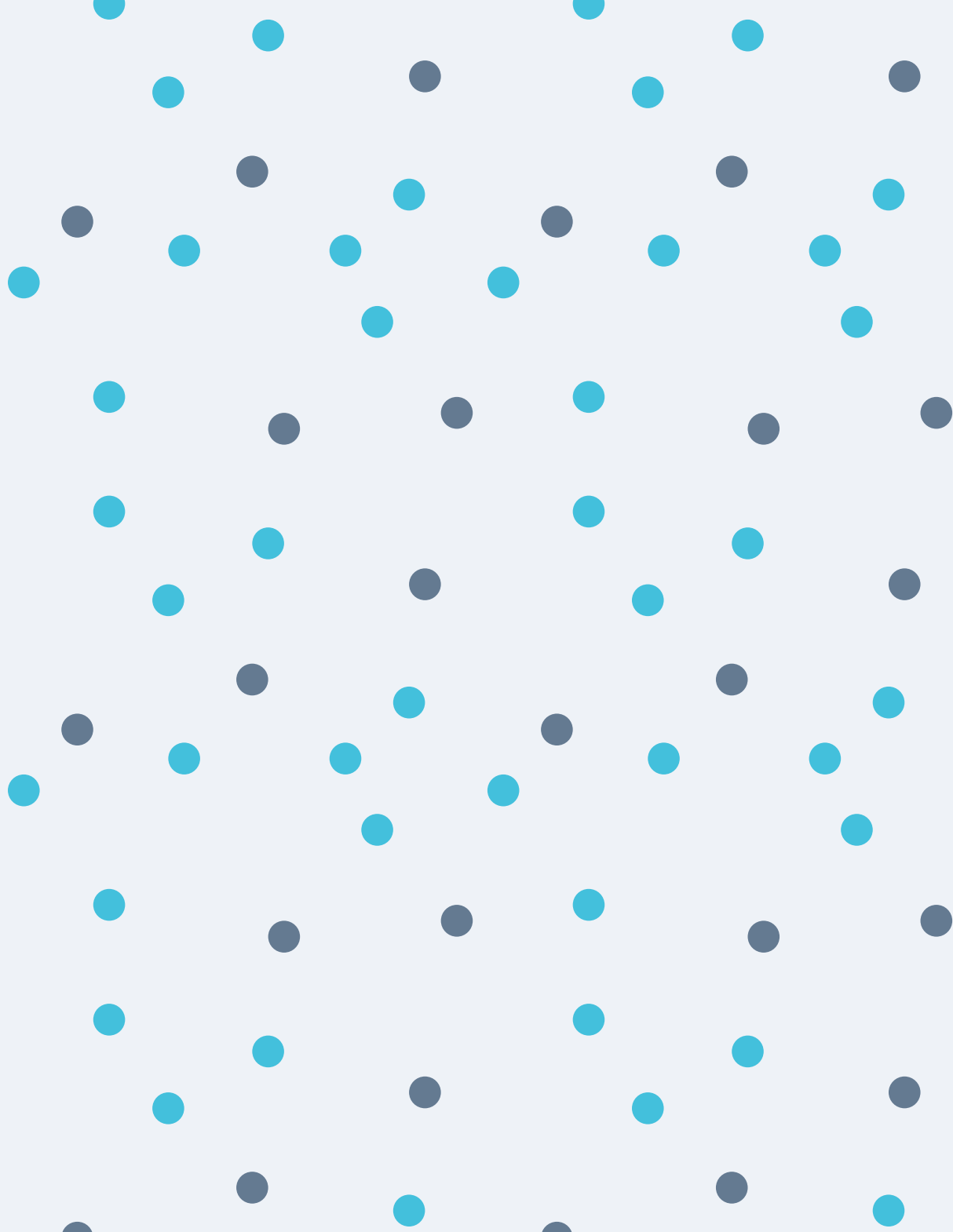


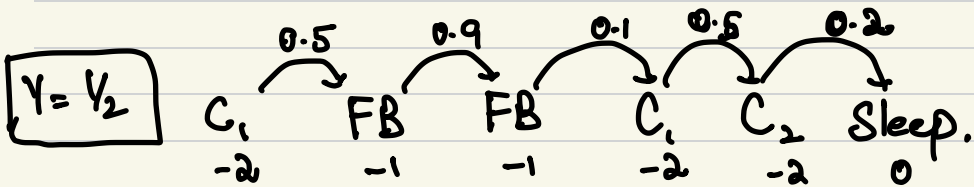


MDP

Return $G_t = R_{t+1} + \gamma R_{t+2} + \dots + \gamma^{T-t} R_{t+T}$

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

$$V(s) = E_{\pi} [G_t | S_t = s]$$



$$V(C_1) = -2 - 1 \times \frac{1}{2} - 1 \times \left(\frac{1}{2}\right)^2 - 2 \times \left(\frac{1}{2}\right)^3$$

$$- 2 \times \left(\frac{1}{2}\right)^4$$

$$= -2 - \frac{1}{2} - \frac{1}{4} - \frac{1}{4} - \frac{1}{8}$$

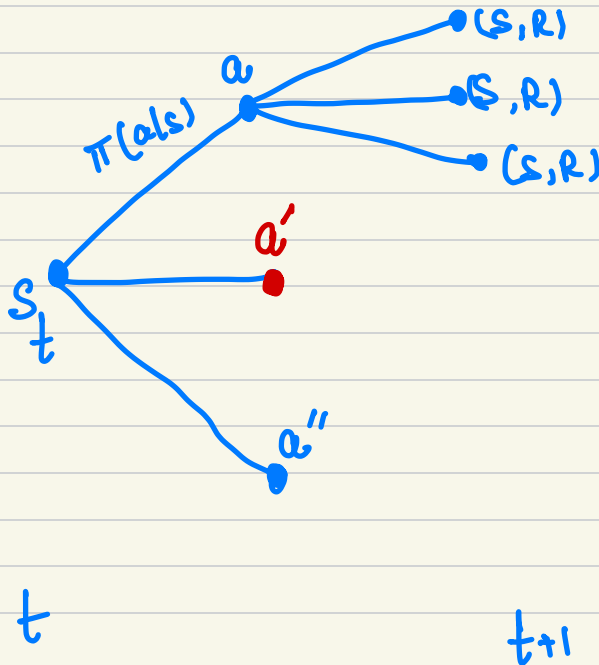
$$= -3 - \frac{1}{8} = -\frac{25}{8} = -3.125.$$

$$V(s) = E [G_t | S_t = s]$$

$$V(s) = E [R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots | S_t = s]$$

$$V(s) = E [R_{t+1} + \gamma [R_{t+2} + \gamma R_{t+3} + \dots] | S_t = s]$$

$$V(s) = E [R_{t+1} + \gamma V(S_{t+1}) | S_t = s]$$



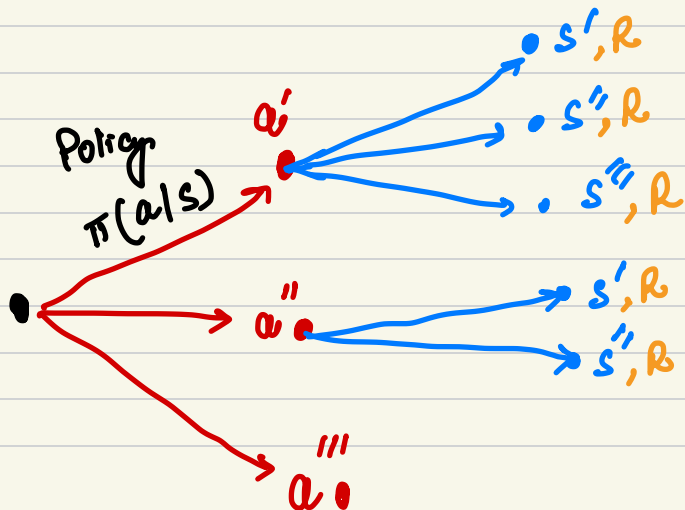
$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$$

$$G_t = R_{t+1} + \gamma (R_{t+2} + \gamma R_{t+3} + \dots)$$

$$\boxed{G_t = R_{t+1} + \gamma G_{t+1}}$$

Now $V(s) = E [G_t | S_t = s]$

$$V(s) = E [R_{t+1} + \gamma G_{t+1} | S_t = s]$$



$$\pi(a_1|s) \left[\begin{array}{l} P(s', R | s, a_1) R \\ + P(s'', R | s, a_1) R \\ + P(s''', R | s, a_1) R \end{array} \right]$$

$$+ \pi(a_2|s) \left[\sum_{s', R} P(s', R | s, a_2) R \right]$$

$$+ \pi(a_3|s) \left[\right]$$

$$= \pi(a_1|s) \left[\sum_{s', R} P(s', R | s, a_1) R \right]$$

$$+ \pi(a_2|s) \sum_{s', R} P(s', R | s, a_2) R$$

$$+ \pi(a_3|s) \sum_{s', R} P(s', R | s, a_3) R$$

$$v_{\pi}(s) = \sum_a \pi(a|s) P(s', R | s, a) [R + \gamma v_{\pi}(s')]$$

$$v^*(s) = \max_{\pi} v_{\pi}(s)$$

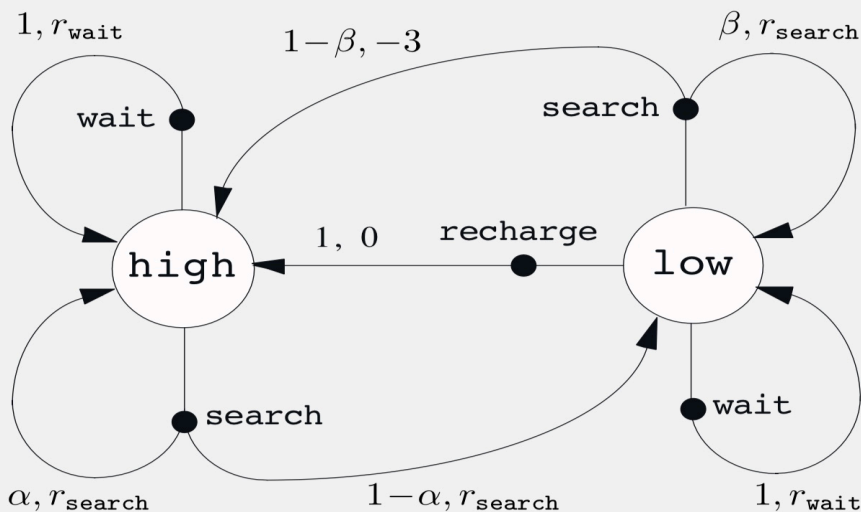
$$q^*(s, a) = \max_{\pi} q_{\pi}(s, a)$$

$$q^*(s, a) = E[R_{t+1} + \gamma v^*(s') | s, a]$$

$$v^*(s) = \max_a q^*(s, a)$$

$$= \max_a E[R_{t+1} + \gamma v^*(s') | s, a]$$

$$= \max_a \sum_s P(s', R | s, a) [R + \gamma v^*(s')]$$



$$a = \{w, r, s\} \quad s = \{h, l\}$$

$$\pi(w|h) = 1 \quad R = r_{\text{wait}}$$

$$\pi(s|h) = \alpha \quad R = r_{\text{search}}$$

$$\pi(s|h) = 1-\alpha \quad R = r_{\text{search}}$$

$$\pi(s|l) = 1-\beta \quad R = -3$$

$$\pi(s|l) = \beta \quad R = r_{\text{search}}$$

$$\pi(r|l) = 1 \quad R = 0$$

$$\pi(w|l) = 1 \quad R = r_{\text{wait}}$$

$$v^*(h) = \max \left[\begin{aligned} & p(h, r | h, w) [r_{\text{wait}} + \gamma v^*(h)] \\ & p(h, r | h, s) [r_{\text{search}} + \gamma v^*(h)] \\ & + p(l, r | h, s) [r_{\text{search}} + \gamma v^*(l)] \end{aligned} \right]$$

$$= \max \left[\begin{aligned} & r_{\text{wait}} + \gamma v^*(h) \\ & \alpha [r_{\text{search}} + \gamma v^*(h)] \\ & + (1-\alpha) [r_{\text{search}} + \gamma v^*(l)] \end{aligned} \right]$$

$$= \max \left[\begin{aligned} & r_{\text{wait}} + \gamma v^*(h) \\ & r_{\text{search}} + \alpha \gamma v^*(h) \\ & + (1-\alpha) \gamma v^*(l) \end{aligned} \right]$$

Bellman equation is:-

$$V(s) = \sum_a \pi(a|s) \sum_{s', R} p(s', R | s, a) [R + \gamma V(s')]$$

or we can write

$$s \quad a \rightarrow a \quad \boxed{R_t = -1}$$

	1	2	3
4	5	6	7
8	9	10	11
12	13	14	

$$= q(1, r) = -1 + V(2) = -1$$

$$S \quad q(1, l) = -1 + V(t) = -1$$

$$q(1, u) = -1 + V(1) = -1$$

$$q(1, d) = -1 + V(5) = -1.$$

	-1	-1	-1
-1	-1	-1	-1
-1	-1	-1	-1
-1	-1	-1	-1

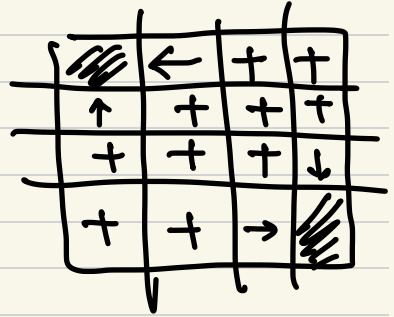
$$V = 0.25 (-1) \times 4$$

$$\boxed{V = -1}$$

	-1	-1	-1
-1	-1	-1	-1
-1	-1	-1	-1
-1	-1	-1	-1

$$\begin{aligned}
 q(1,8) &= -1 + v(2) \\
 &= -1 + (-1) \\
 &= -2
 \end{aligned}$$

$$q(1,8) = -2$$



$$q(1,u) = -1 + v(1)$$

$$q(1,u) = -2$$

$$q(1,l) = -1 + v(t)$$

$$q(1,l) = -1$$

$$q(1,d) = -1 + v(5)$$

$$q(1,d) = -2$$

$$q(1,1) = -1 + 0 = -1.$$

$$v(1) = -1$$

Ms -2025 -23 Awaraz Ali

25- 12 Usama.

25- 15 Saad

24 Hassan Farooq

24- 13 Faraz Hassan
Ali

-20 Zohaib Ahmad

25- 03 M. Umer

25- 09 Faisal

25- 13 Fathu

25- 19 Salim

25-28 Romana

24-02 Wareeha

25-08 Wajeelha

25-22 Muneeza

25-01 Momina

25-05 Aqsa

25-11 Hassan Ali

Attendance

11/02

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Section. B

11/02.

359	358	338	354
360	332	339	328
345	348	351	335
341	363	355	
362	331	346	
353	334	350	
364.	352		
	329		
321	319	327	330
317	315	347	361
311	313	344	342
326	316	333	318
322	324	365	320
	323	340	312
		357	

(B)

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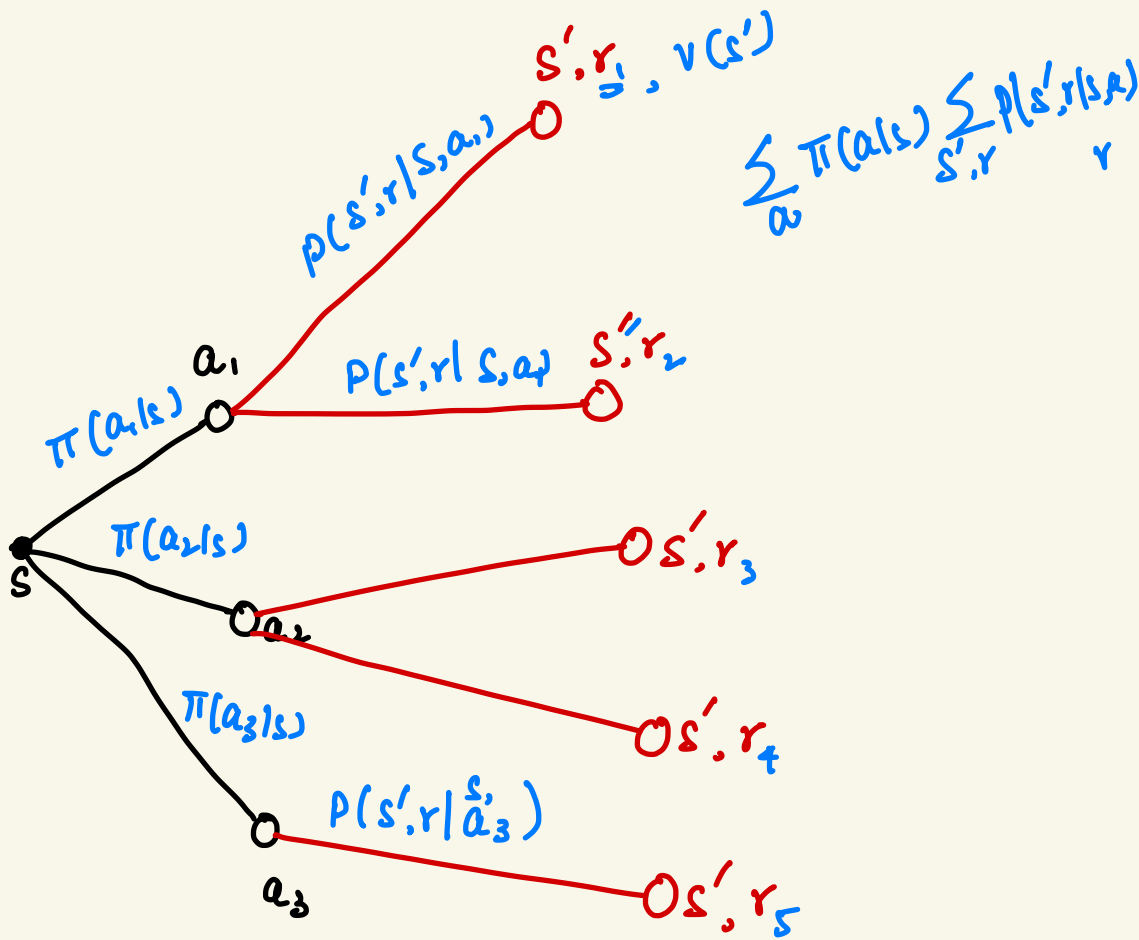
Bellman Eq.

$$V_{\pi}(s) = \sum_a \pi(a|s) \sum_{s', r} p(s', r|s, a) [r + \gamma V_{\pi}(s')]$$

$$R(s) = \sum_a \pi(a|s) \sum_{s', r} p(s', r|s, a) r$$

Reward Function

$$V_{\pi}(s) = R(s) + \gamma [p(s' | s) V_{\pi}(s')]$$



$$\pi(a_1|s) \left[p(s', r_1 | s, a_1) v_1 + p(s'', r_2 | s, a_1) v_2 \right]$$

$$+ \pi(a_2|s) \left[p(s', r_3 | s, a_2) v_3 + p(s', r_4 | s, a_2) v_4 \right]$$

$$= \pi(a_1|s) \sum_{s', r} p(s', r | s, a_1) v + \pi(a_2|s) \sum_{s', r} p(s', r | s, a_2) v$$

Q1:- what is the diff b/w
reward and return?

Q2:- what is MDP?

Q3:- what is a value function?
what is the diff b/w
state and action value
function?

Q4:- what are episodic Tasks?

Q5:- No of steps in each episode of an episodic task is fixed.

Q6:- why we use discount factor?

Q7:- in a continuing task, reward is always ± 1 . Discount factor is 0.9. How to find the return?

$$\begin{aligned} & 1 + 0.9(1) + 0.9^2(1) + \dots \\ & (0.9)^0 + (0.9)^1 + (0.9)^2 + \dots \\ & \sum_{k=0}^{\infty} 0.9^k \\ & \frac{1}{1-0.9} = 10 \end{aligned}$$

Q8 Can you explain a deterministic and a stochastic policy?

Q9:- Can you explain how episodes are generated?

	a_1	a_2	a_3	a_4	a_5
s_1	0.3	0.3	0.3	0.1	0
s_2					
s_3					

$\pi(a|s)$

378

Always go slow

$$v(s) = \begin{bmatrix} \frac{c}{0} & \frac{w}{0} & \frac{0}{0} \end{bmatrix} \quad \gamma = 0.5$$

↳ terminal 0

$$Q(c, s) = p(c, +1 | c, s) [1 + (0.9) v(c)]$$

$$= 1 [1] = 1.$$

$v(c) =$

$$Q(c, F) = p(w, +2 | c, F) [2 + (0.9) v(w)]$$

$$+ p(c, +2 | c, F) [2 + (0.9) v(c)]$$

$$= 0.5(2) + 0.5(2)$$

$$Q(c, F) = 2.$$

$$v(c) = 0.5 \times 1 + 0.5 \times 2$$

$$v(c) = 0.5 + 1 = 1.5$$

$$Q(w, s) = 0.5 [1 + 0.5 v(c)] + 0.5 [1 + 0.5 v(w)]$$

$$= 0.5 + 0.5 = 1.$$

$$v = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix}$$

$$v(c) = 1 \left[1 + 0.5(1) \right] = 1.5$$

$$v(w) = 0.5 \left[1 + 0.5(1) \right] + 0.5 \left[1 + 0.5 \right] = 1.5$$

$$v = \begin{bmatrix} 0.5 & 0.5 & 0 \end{bmatrix}$$

$$v = \begin{bmatrix} 1.5 & 1.5 & 0 \end{bmatrix}$$

$$v(c) = 1 \left[1 + 0.5(1.5) \right] = 1.75$$

$$v(w) = 0.5 \left[1 + 0.5(1.5) \right] + 0.5 \left[1 + 0.5(1.5) \right] = 1.75$$

$$v = \begin{bmatrix} 1.75 & 1.75 & 0 \end{bmatrix}$$

$$v(c) = 1 \left[1 + 0.5(1.75) \right] =$$

$$v(s) = 2$$

$$v(w) = 2$$

$$v(o) = 0.$$

$$\pi(c) = \max \left[\begin{array}{l} \text{S} \\ \text{F} \end{array} \right] \begin{array}{l} 1[1 + 0.5(2)] = 2 \\ 0.5[2 + 0.5(2)] \\ + 0.5[2 + 0.5(2)] \end{array} = 3$$

$$\pi(c) = \text{fast}$$

$$\text{fast, slow}$$